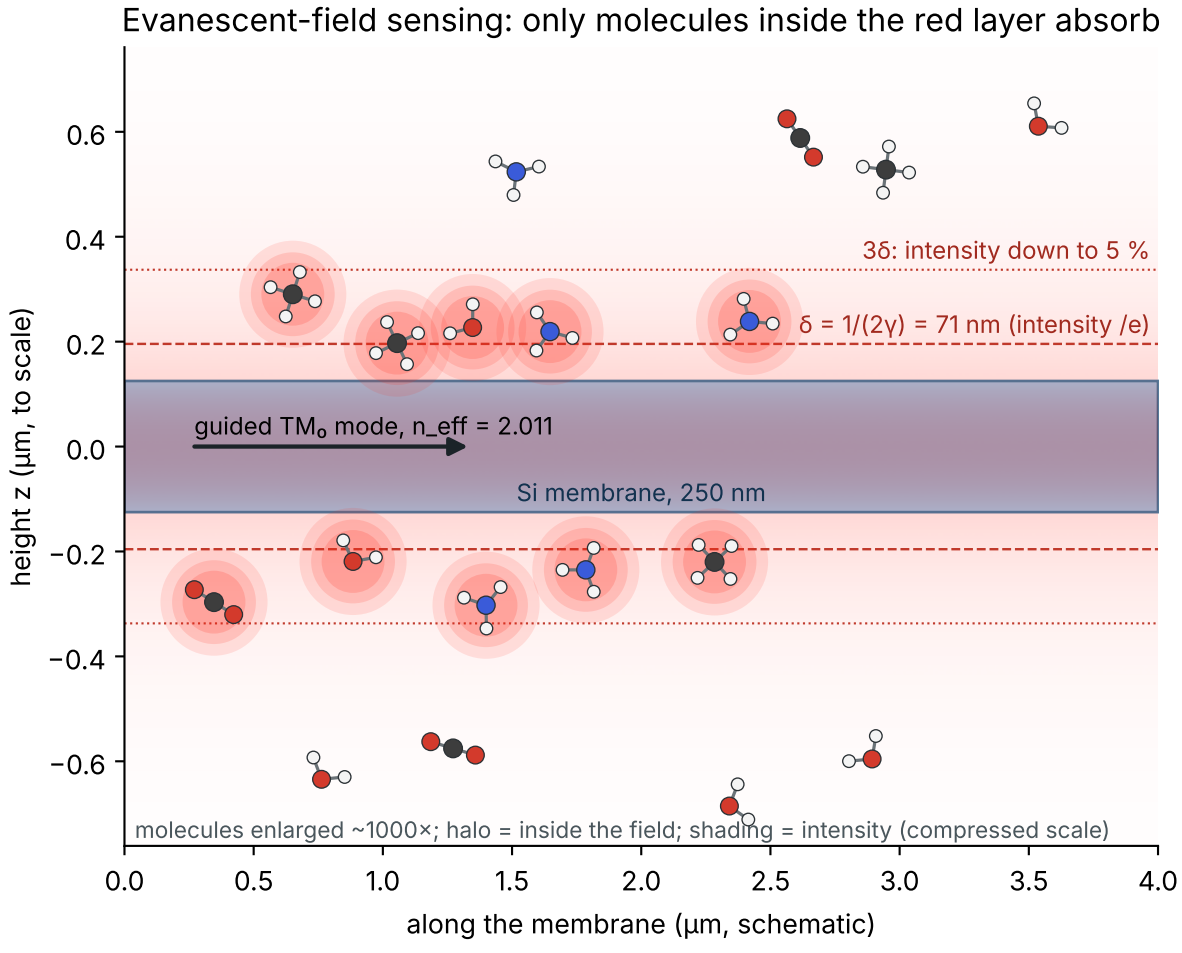
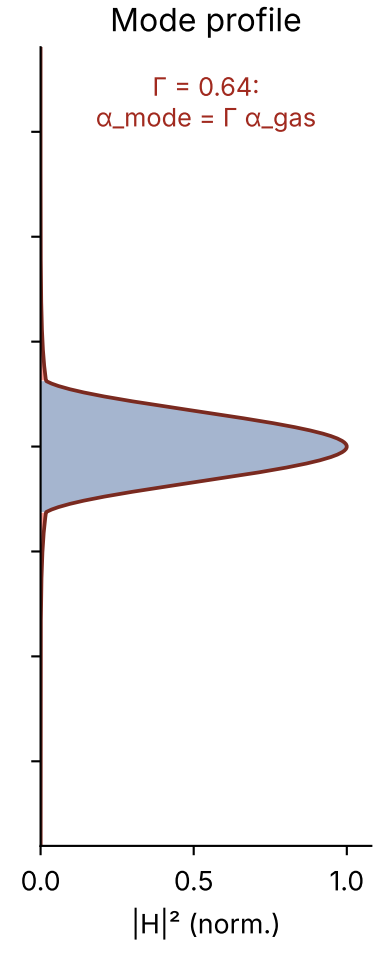
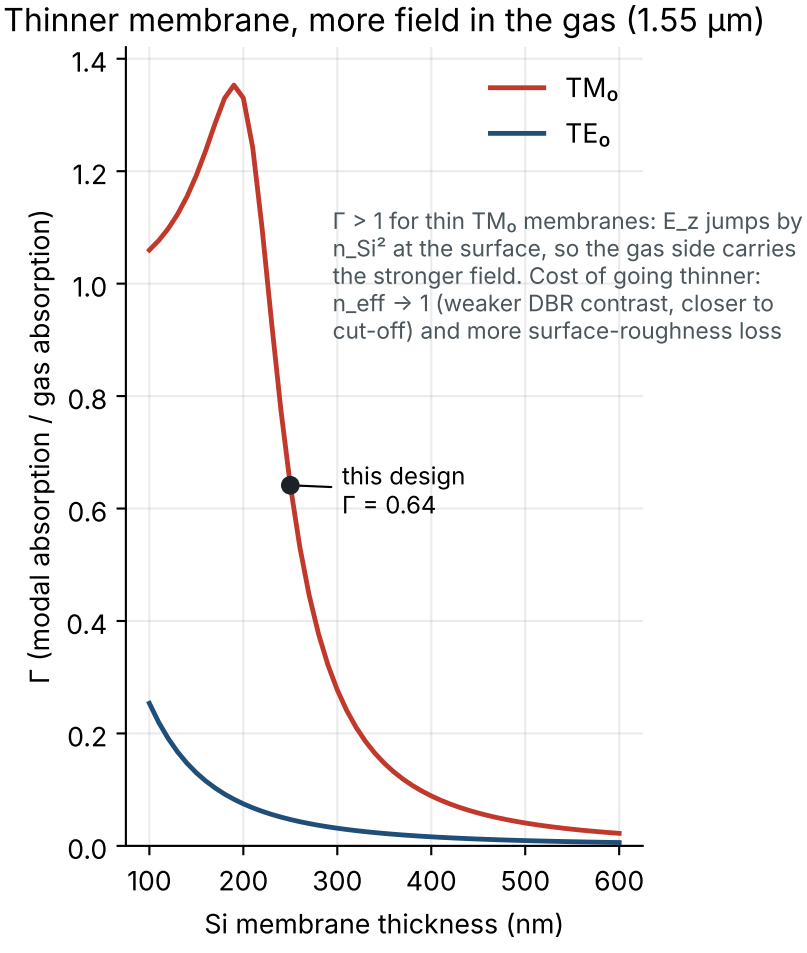

Evanescent-field sensing: only molecules inside the red layer absorb
3δ: intensity down to 5 %
δ = 1/(2γ) = 71 nm (intensity /e)
guided TM₀ mode, n_eff = 2.011
Si membrane, 250 nm
molecules enlarged ~1000×; halo = inside the field; shading = intensity (compressed scale)
height z (µm, to scale)
along the membrane (µm, schematic)


Mode profile
Γ = 0.64:
α_mode = Γ α_gas
|H|² (norm.)


Thinner membrane, more field in the gas (1.55 µm)
TM₀
TE₀
Γ > 1 for thin TM₀ membranes: E_z jumps by n_Si² at the surface, so the gas side carries the stronger field. Cost of going thinner: n_eff → 1 (weaker DBR contrast, closer to cut-off) and more surface-roughness loss
this design
Γ = 0.64
Γ (modal absorption / gas absorption)
Si membrane thickness (nm)